

## **Module – backup and restore**

1. Create a backup of your Music folder (or any folder with at least 3 files) using the tar command and compress it with gzip, saving the output as music\_backup.tar.gz in your home directory.

### **ANS :**

#### ❖ **Create a compressed tar.gz backup:**

- tar -czvf ~/music\_backup.tar.gz ~/Music

#### ❖ **Verify the backup file:**

- ls -lh ~/music\_backup.tar.gz

- A compressed backup named music\_backup.tar.gz was created in the home directory.

The archive contains all files and subdirectories from the Music folder.

2. Extract the contents of a tar.gz file (for example, music\_backup.tar.gz) into a new directory called restored\_music and verify that all files are restored correctly.

### **ANS :**

#### ❖ **Create a new directory:**

- mkdir ~/restored\_music

#### ❖ **Extract the archive:**

- tar -xzf ~/music\_backup.tar.gz -C ~/restored\_music

#### ❖ **Verify restored files:**

- ls -R ~/restored\_music

- The archive was extracted successfully into restored\_music.  
All original files and folders were restored correctly.

3. Write a shell command that creates a compressed backup of all .jpg images from your Downloads folder into a file named images\_backup.tar.gz, excluding any files larger than 5MB.  
  
**Hint:** Use the find command with size filtering and tar with --files-from.

### **ANS :**

#### ❖ **Command:**

- find ~/Downloads -type f -name "\*.jpg" -size -5M > jpg\_list.txt  
tar -czvf ~/images\_backup.tar.gz -T jpg\_list.txt

#### ❖ **Verify Archive:**

- tar -tzvf ~/images\_backup.tar.gz

- A compressed archive named images\_backup.tar.gz was created.  
Only .jpg files smaller than 5MB were included in the backup.

4. Simulate a restore scenario: delete one file from your original Music folder, then use your music\_backup.tar.gz archive to restore only the deleted file without overwriting the rest.

ANS :

- ❖ Assume the file song1.mp3 was deleted:
  - `rm ~/Music/song1.mp3`
- ❖ View files inside archive:
  - `tar -tzvf ~/music_backup.tar.gz`
- ❖ Restore only the deleted file:
  - `tar -xzvf ~/music_backup.tar.gz Music/song1.mp3 -C ~/`
- ❖ Verify restoration:
  - `ls ~/Music`
- The deleted file song1.mp3 was restored successfully from the backup archive. No other files in the Music folder were modified or overwritten.